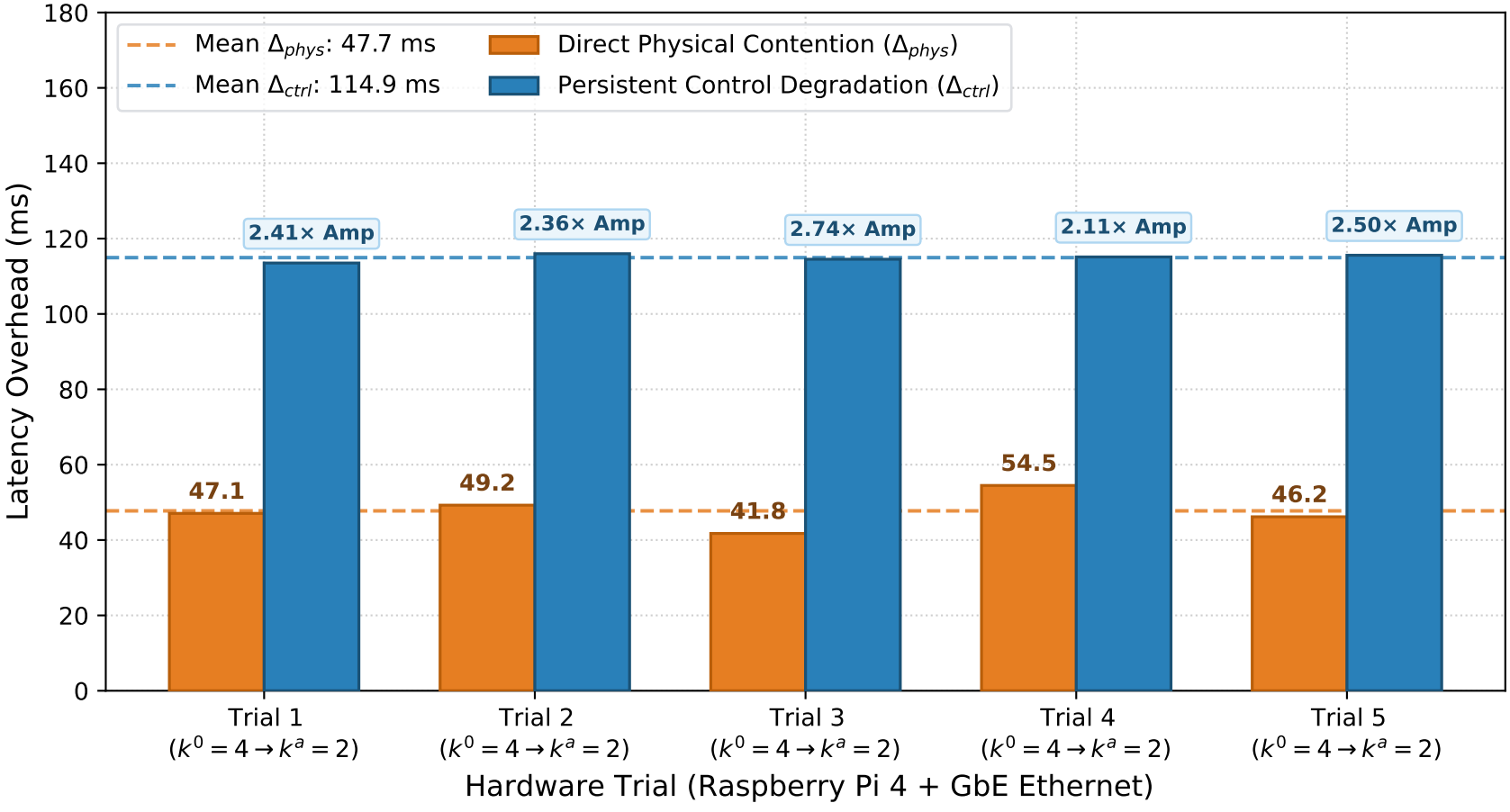
Physical Contention (Δ_{phys}) vs. Control Amplification (Δ_{ctrl}) on Real Hardware
Mean Δ_{phys}: 47.7 ms
Mean Δ_{ctrl}: 114.9 ms
Direct Physical Contention (Δ_{phys})
Persistent Control Degradation (Δ_{ctrl})
Latency Overhead (ms)
0
20
40
60
80
100
120
140
160
180
2.41× Amp
2.36× Amp
2.74× Amp
2.11× Amp
2.50× Amp
47.1
49.2
41.8
54.5
46.2
Trial 1 ($k^0 = 4 \rightarrow k^a = 2$)
Trial 2 ($k^0 = 4 \rightarrow k^a = 2$)
Trial 3 ($k^0 = 4 \rightarrow k^a = 2$)
Trial 4 ($k^0 = 4 \rightarrow k^a = 2$)
Trial 5 ($k^0 = 4 \rightarrow k^a = 2$)
Hardware Trial (Raspberry Pi 4 + GbE Ethernet)